

MIMIC-III Recurrent-Event EDA

Initial exploratory analysis of the 35,643-admission cohort in `processed-data/input.csv` , set up for the recurrent-event readmission model. Companion script: `mimic_analysis/testing/eda_recurrent.py` .

1. Cohort layout

- `input.csv` has **one row per patient**: 35,643 rows ↔ 35,643 unique `SUBJECT_ID` s. Each row is the patient's **indexed admission** (the one with the V1..V26 covariates).
- To see the recurrent-event structure we pull **all** admissions for these patients from `raw-data/ADMISSIONS.csv` :

Quantity	Value
Patients (= rows in <code>input.csv</code>)	35,643
Total admissions across all visits	46,263
Readmission events (= admissions – 1)	10,620
Mean events / patient	0.30
Median events / patient	0
Patients with zero events (single admission)	29,284 (82.2%)
Max admissions for one patient	42
Patients whose last admission ended in death	4,723 (13.3%)

Admissions-per-patient histogram (head):

n_admissions	patients
1	29,284
2	4,305
3	1,153
4	442
5	221
6	94
7	43
8	25
9	23
10	14
...	...

Follow-up days: median 8.5, mean 131, max 4,129 (~11.3 yr). A handful of negative values exist from rare `DISCHTIME < ADMITTIME` records in MIMIC.

2. Missingness in V1..V26

Tier	Columns	% missing	Comment
High	<code>PaO2FiO2_vent_min</code>	58.2%	only defined on ventilated patients with an ABG
High	<code>bilirubin_min</code> , <code>bilirubin_max</code>	56.4%	non-routine lab, ordered on clinical suspicion
Medium	<code>tempc_min</code> , <code>tempc_max</code>	1.9%	sporadic non-recording
Medium	<code>urineoutput</code>	1.6%	requires urinary catheter or charted output
Low	<code>wbc</code> , <code>bicarbonate</code> , <code>bun</code> , <code>sodium</code> , <code>sysbp</code> , <code>gcs</code> , <code>heartrate</code> , <code>potassium</code>	0.2–0.9%	almost always collected
Zero	<code>age</code> , <code>AIDS</code> , <code>HEM</code> , <code>METS</code> , <code>Adm_type</code>	0%	demographic / categorical / always set

What `PaO2FiO2_vent_min` actually is

The **PaO₂/FiO₂ ratio** divides arterial oxygen partial pressure (PaO₂ , mmHg, from an arterial blood gas) by the fraction of inspired oxygen (FiO₂ , 0.21–1.0, from the ventilator setting). It is the standard bedside measure of pulmonary gas-exchange impairment and is the defining axis of the Berlin ARDS criteria:

PaO ₂ /FiO ₂	Severity
≤ 300	mild ARDS
≤ 200	moderate ARDS
≤ 100	severe ARDS

The `_vent_min` suffix means *minimum* value observed **while the patient was mechanically ventilated** during the first-24-hour window. A missing value typically means the patient was **not ventilated**, or was ventilated but had no arterial blood gas drawn — both signals of a less-acute respiratory state. (Conversely, a low recorded value is a strong marker of severe respiratory failure.)

What `bilirubin_min/max` actually are

Serum **total bilirubin** (mg/dL), a marker of hepatobiliary function (liver disease, biliary obstruction) and red-cell breakdown (hemolysis). Normal range is roughly 0.1–1.2 mg/dL. Elevated bilirubin is a component of:

- **SOFA** liver subscore (used for organ-failure tracking)
- **MELD** (end-stage liver disease score)
- **APACHE / SAPS** severity scores

Bilirubin is **not** part of a standard daily metabolic panel — it is ordered when there is clinical suspicion of liver dysfunction, jaundice work-up, sepsis screening with hepatic concern, or post-operative monitoring after hepatobiliary surgery. The 56% missingness reflects this selective ordering pattern.

3. "Tested vs not-tested" as an informative signal

For several features in this dataset, **missingness is itself informative** — the act of ordering or not ordering a test reflects the clinician's prior judgment of the patient's clinical state. This is especially true for:

Column	Why missingness is informative
<code>PaO2FiO2_vent_min</code>	Missing ⇔ patient not ventilated / no ABG ⇔ less acute respiratory failure (a <i>protective</i> signal). Recorded ⇔ ICU-level respiratory support, often associated with worse prognosis.
<code>bilirubin_min/max</code>	Missing ⇔ no clinical reason to suspect liver involvement. Recorded ⇔ clinician triggered a liver work-up (often higher acuity).
<code>urineoutput</code>	Missing ⇔ no indwelling urinary catheter, often less critically ill. Recorded ⇔ catheterized, typically ICU-level care.
<code>tempc</code>	Missing ⇔ patient not actively monitored on telemetry/ICU floor at the time.

Why this matters for the model

The standard practice of "impute the median and proceed" **discards the missingness signal** and can bias the recurrent-event hazard estimates because the missingness mechanism is correlated with the outcome (readmission and survival). Two reasonable mitigations:

1. **Missingness-indicator augmentation.** For each high-missingness feature add a binary `M_j = 1{X_j is missing}` column and impute the underlying value with the cohort mean (or any neutral value). The indicator absorbs the informative-missingness signal cleanly and lets the regression separate "what the test value was" from "whether the test was ordered." Best-suited to:
2. `PaO2FiO2_vent_min` → `M_vent`
3. `bilirubin_min/max` → `M_bilirubin`
4. **Joint missingness model (MNAR).** When the missingness mechanism itself depends on outcome, a joint model of the covariate process and the readmission process is more principled but considerably heavier to fit.

For the **informative censoring** problem (some patients are censored because they died, and death is correlated with the X covariates), indicator-based handling of missingness pairs naturally with a frailty term that ties the censoring time to the recurrent-event process.

4. Correlation across V1..V26 ($|r| \geq 0.6$)

All six high-correlation pairs are min/max readings of the same lab within the 24-hour window — i.e., the spread between the two summaries is small for that lab.

Pair	r
<code>bilirubin_min</code> / <code>bilirubin_max</code>	+0.983
<code>bun_min</code> / <code>bun_max</code>	+0.958
<code>wbc_min</code> / <code>wbc_max</code>	+0.873

Pair	r
bicarbonate_min / bicarbonate_max	+0.810
sodium_min / sodium_max	+0.705
heartrate_max / heartrate_min	+0.605

The first four are effectively collinear. Two ways to handle this:

- Replace each (X_min, X_max) pair with (mean, range) — same information, lower correlation.
- Drop one side per pair and keep only the more clinically informative summary (typically the *abnormal-direction* extreme: e.g., keep bilirubin_max, bun_max, wbc_max, drop the mins).

sodium and heartrate are borderline and probably fine to keep as min/max.

5. Data-quality flags worth knowing

- **age max = 308.97**: MIMIC adds approximately 300 years to the DOB of any patient ≥ 89 at admission for de-identification. These rows should be re-coded to a fixed value (e.g., 90) before fitting, not left at face value.
- **urineoutput** : 6 negative values, max 98,280 mL — both are implausible and suggest sentinel/entry errors. Clip at clinically plausible bounds.
- **PaO2FiO2_vent_min max = 35,400**: physiologically implausible (normal upper bound ≈ 500). Likely a unit/charting error in a few rows; clip or drop.
- A few DISCHTIME < ADMITTIME records in ADMISSIONS.csv produce negative follow-up days at the patient level. Either drop those patients or set their follow-up to 0.

6. Verification of cohort assumptions

Companion script: `mimic_analysis/testing/verify_first_admission.py`.

Is the indexed admission the patient's first hospital admission?

The PDF says `valid_aids.csv` covers patients alive 24h after their **first ICU admission**. Empirically:

Check	Patients	%
Indexed HADM_ID == patient's first hospital admit	33,677	94.5%
Indexed HADM_ID != first hospital admit (had earlier non-ICU stay)	1,966	5.5%

For the 1,966 with prior non-ICU admissions, the median gap between first hospital admit and first ICU admit is **309 days**. The X covariates in `input.csv` describe the first ICU stay, *not* the very first hospitalization. Two options:

- Keep them, treat the cohort as "first ICU stay" (current default; we add a `is_first_hospital_admit` flag column).
- Drop them for clean "X at first hospital admission" semantics.

DISCHTIME < ADMITTIME records

Where	Count	Death rate (HOSPITAL_EXPIRE_FLAG = 1)
All of ADMISSIONS.csv	98	80 (81.6%)
In our cohort	0	—

The 98 globally-bad records are dominantly in-hospital deaths with mangled timestamps; **none are in our cohort**. However, **4 cohort patients** still end up with negative *patient-level* follow-up (their last DISCHTIME across all admissions is earlier than the first ADMITTIME). All 4 died on their first admission → safe to drop as data errors.

Are X features available at first admission?

- **From the PDF**: `input.csv` is "non-temporal summary statistics within the first 24 hours after admission." `valid_aids.csv` patients are alive 24h after first ICU admit. So X is, by construction, available at the index admission and contains no per-event timestamps (it's already collapsed to one row per patient).
- The richer time-resolved version (24 hourly bins $\times$ 15 vars) lives in `imputed-normed-ep_1_24.npz` (`ep_tdata`).

7. Cohort cleaning (`clean_cohort.py`)

Companion script: `mimic_analysis/clean_cohort.py`. Outputs `merged_data/cohort_clean.csv` (35,639 $\times$ 37). Each cleaning step prints what it changed; summary below.

What was cleaned

Step	Rows / values affected	Action
Age cap at 90 (MIMIC ~300 yr offset for ≥ 89)	1,809 patients (5.1%)	<code>age = 90</code> ; flag in <code>age_capped</code>
Clip X to clinical bounds	287 values across 12 cols	<code>clip(lo, hi)</code> per column
Missingness indicators	3 cols (PaO2FiO2, bilirubin_min/max)	added <code>M_<col></code> binary cols
Drop patients with <code>DISCHTIME < ADMITTIME</code> on any admission	13 patients	removed (unreliable timestamps)
Drop residual negative follow-up	4 patients	removed
Flag indexed-vs-first-hospital-admit	1,966 patients (5.5%)	added <code>is_first_hospital_admit</code>

Final cohort after both drops: **35,626 patients**.

Bounds used (only the clipped columns shown):

Column	(lo, hi)	n below lo	n above hi
PaO2FiO2_vent_min	(50, 600)	73	39
urineoutput	(0, 20000)	6	7
tempc_min	(25, 45)	5	0
sysbp_min	(30, 300)	91	0
sysbp_max	(30, 300)	0	2
heartrate_min	(20, 250)	18	0
heartrate_max	(20, 250)	0	1
wbc_max	(0, 500)	0	3
potassium_min	(1, 10)	5	0
potassium_max	(1, 10)	0	22
sodium_min	(100, 180)	12	0
sodium_max	(100, 180)	0	3

Bilirubin and BUN bounds were specified but no values fell outside — the data is well-behaved there.

Final cohort: 35,626 patients

Statistic	Value
Patients	35,626
Mean events / patient	0.298
Patients with 0 events	29,280 (82.2%)
Patients died at last admit (death-censored)	4,712 (13.2%)
Patients with <code>is_first_hospital_admit = 1</code>	33,666 (94.5%)

Follow-up time distribution

Quantile	days
25%	4.8
50% (median)	8.5
75%	20.3
90%	234.9
95%	903.9
99%	2,443.9
max	4,129 (~11.3 yr)

Heavy right tail. 75% of patients are followed ≤ 20 d, but the top 1% extends past 6.7 years. For numerical stability you may want to cap follow-up at, e.g., 5 years (1,825 d) — this affects only the top ~3-4% of patients but trims the long noisy tail of rare-late events.

Output schema (`cohort_clean.csv` , 37 cols)

```
HADM_ID, SUBJECT_ID,
V1..V26 (cleaned/clipped),
age_capped, M_PaO2FiO2_vent_min, M_bilirubin_min, M_bilirubin_max,
n_admissions, n_events, follow_up_days, died_at_end,
is_first_hospital_admit
```

8. Long-format recurrent-event data (build_long_format.py)

The recurrent-event estimator (`recurrent_ode.fit`) takes long format: one row per event + one censoring row per subject, with the **X** covariate vector repeated on every row.

Modeling convention used here: each patient has **one X vector** (the V1..V26 + indicators from their first admission). They contribute **zero or more event rows** (`delta = 1` , one per *subsequent* admission) plus exactly **one censoring row** (`delta = 0` , the final row).

Time origin and censoring

Quantity	Definition
Time origin (<code>time = 0</code>)	first admission's <code>ADMITTIME</code>
Event time (k-th readmission)	<code>ADMITTIME_k - ADMITTIME_1</code> (days)
Censor time	<code>max(last DEATHTIME, last DISCHTIME, last ADMITTIME) - ADMITTIME_1</code>
<code>is_death_censored</code> flag	1 if <code>last DEATHTIME</code> is non-null (informative censoring)

The fallback to `last ADMITTIME` in the censor-time formula handles admissions with missing `DISCHTIME` (transfers / open episodes), which otherwise can produce a censor row earlier than the last event row. After this fix, **every patient's last row has `delta = 0`**.

Long-format summary

Statistic	Value
Patients (unique <code>id</code>)	35,626
Total rows	46,227
Event rows (<code>delta = 1</code>)	10,601
Censoring rows (<code>delta = 0</code>)	35,626
Rows per patient: min / median / max	1 / 1 / 42
Death-censored patients	4,712 (13.2%)

A patient with no readmissions contributes exactly one row (their censor). A patient with *k* readmissions contributes *k* + 1 rows.

Output schema (cohort_long.csv , 34 cols)

```
id, time, delta,
V1..V26 (constant within id),
age_capped, M_PaO2FiO2_vent_min, M_bilirubin_min, M_bilirubin_max,
is_death_censored
```

Example (`id = 17` , one readmission at day 133, censored at day 137):

```
id      time  delta  age  is_death_censored
17 133.288889    1  47.45                0
17 137.309028    0  47.45                0
```

Format match against `recurrent_ode.fit()`

`recurrent_ode.fit()` (and the underlying `_save_data_npz`) expects:

```
data = {
    'id': np.ndarray,    # shape (n,) or (n,1), int
    'time': np.ndarray,  # shape (n,) or (n,1), float
    'delta': np.ndarray, # shape (n,) or (n,1), int (0 or 1)
    'x': np.ndarray,    # shape (n_rows, p), float
}
```

The build script writes both a CSV (`cohort_long.csv`) and a ready-to-fit NPZ (`cohort_long.npz`). Side-by-side:

Field	Model expects	cohort_long.npz (ours)	Match
<code>id</code>	int (n,)	(46227,) int64	✓

Field	Model expects	cohort_long.npz (ours)	Match
time	float (n,)	(46227,) float64	✓
delta	int 0/1 (n,)	(46227,) int64	✓
x	float (n, p)	(46227, 30) float64	✓

Plus the model assumes:

- Last row per `id` is the censoring row (`delta = 0`) — **OK**
- Times within `id` are non-decreasing — **OK**
- No NaN in `x` — **OK** (NaNs filled with 0; the `M_*` indicator columns retain the "test was/wasn't ordered" signal)

NaN handling

The model does not accept NaN. Each missing entry in `V1..V26` is filled with `0` and the missingness signal is preserved separately by the `M_*` indicator columns. Imputation breakdown for the long-format file:

Column	NaN → 0
PaO2FiO2_vent_min	27,757
bilirubin_min / bilirubin_max	25,385 each
urineoutput	1,073
tempc_min / tempc_max	805 / 803
wbc_min/max , bicarbonate_min/max	380 / 365 each
bun_min/max , sodium_min/max , gcs_min , potassium_min/max	< 200 each

Total imputed values: **83,905** (across 16 columns).

If you'd rather impute with the cohort mean (or fit-fold mean), swap the `np.nan_to_num(x, nan=0.0)` line in `build_long_format.py`.

Loading for the model

```
import numpy as np
from recurrent_ode import fit

d = np.load('mimic_analysis/merged_data/cohort_long.npz')
data = {'id': d['id'], 'time': d['time'],
        'delta': d['delta'], 'x': d['x']}

est = fit(data, model='aft', knots='quantile', ci=True)
print(est.beta)
```

9. Outputs

File (under <code>merged_data/</code>)	What
missingness.csv	Per-column NaN / zero / negative / min / max
corr_matrix.csv	Full 26 × 26 Pearson correlation matrix
per_patient_summary.csv	One row per patient: <code>n_admissions</code> , <code>n_events</code> , <code>follow_up_days</code> , <code>died_at_end</code>
cohort_clean.csv	35,626 × 37 — wide format, one row per patient
cohort_long.csv	46,227 × 34 — long format, human-readable
cohort_long.npz	dict with <code>id / time / delta / x</code> — ready for <code>recurrent_ode.fit()</code>

Reproduce with:

```
python3 mimic_analysis/testing/eda_recurrent.py      # raw EDA tables
python3 mimic_analysis/testing/verify_first_admission.py  # cohort checks
python3 mimic_analysis/clean_cohort.py              # cleaned wide cohort
python3 mimic_analysis/build_long_format.py          # long-format input
```